

OraMet

SIH 2026 | Problem Statement 26192 | Disaster Management

- **Android-first prototype for local situational awareness during flood emergencies in hilly communities.**
- **Project/app name: OraMet. Registered team name, team ID, institute, and member names must be added from the SIH portal.**
- **Prototype demonstration - not an operational early-warning or emergency-dispatch service.**

The problem and our design goal

A useful interface must make uncertainty visible

- **Residents need understandable local context, nearby mapped places, and quick access to emergency actions.**
- **A prototype without authorized river, shelter, and road data must not invent certainty or label a place safe.**
- **OraMet brings available location/weather context and manual emergency tools into one Android experience.**

What works in the prototype today

Android-first React Native application

- **Device location permission flow and approximate area selection.**
- **Open-Meteo current-weather lookup when reachable; app thresholds are heuristic only.**
- **OpenStreetMap/Overpass nearby mapped-place lookup with unverified status and straight-line distance.**
- **Local persistence, emergency guidance, 112 calling action, and a user-reviewed SMS draft.**

A transparent resident workflow

Location -> context -> options -> user action

- **Choose precise device location or an approximate area; approximate points are not used for local risk scoring or place sharing.**
- **View current weather only when a valid response is received; otherwise the app shows unavailable.**
- **Review nearby mapped places with an explicit notice: not authority-verified, safe, or confirmed open.**
- **Open generic map directions only after a warning; directions are not hazard-aware.**
- **Prepare an SMS draft or call 112 directly. User reviews and sends; delivery is not confirmed.**

Data boundaries are explicit

No fabricated official readings or all-clear claims

- **No authorized IMD, CWC river-gauge, GSI landslide, NHAI road-closure, ISRO/MOSDAC, or WMO forecast feed is configured.**
- **No validated flood-crest lead-time prediction, verified shelter directory, or safe evacuation route is available.**
- **No automatic shake detection, audible siren, background push, SMS delivery receipt, or 112/CAD dispatch integration.**
- **Open-Meteo and OpenStreetMap are public services; neither is an emergency authority.**

Validation and next step

Prototype validated; field deployment requires partnerships

- **TypeScript, lint, 8 tests, and Android Metro JavaScript bundle pass.**
- **Native Android Gradle build was not run in this environment because Java/JDK is unavailable.**
- **Next: agency approvals and data agreements; verified shelter/road data; secure backend and identity; approved alert delivery; field and accessibility validation.**
- **Operational use requires authority integration, security/privacy review, hosting, maintenance, and responder agreements.**